Lesson 01 Demo 04

Performing CRUD Methods in a JSON Database

Objective: To perform create, read, update, and delete (CRUD) operations on a JSON database using various HTTP methods and endpoints with curl

Tools required: Visual Studio Code and json-server

Prerequisites: Refer to Lesson_01_Demo_01_Creating_and_Using_a _JSON_Server_File

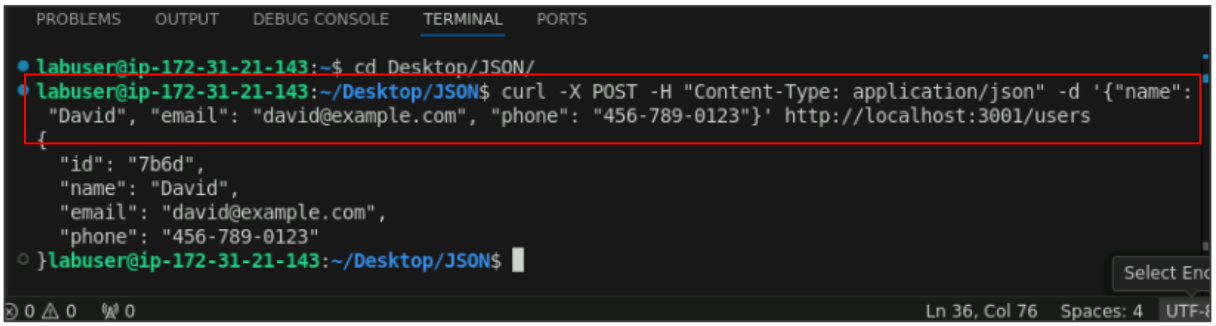
Steps to be followed:

1. Create a new user in the users collection
2. Retrieve data for a specific user (by ID)
3. Update data for a specific user (by ID)
4. Delete a specific user (by ID)

Step 1: Create a new user in the users collection

1.1 Open the terminal and run the following curl command to create a new user named David in the database:

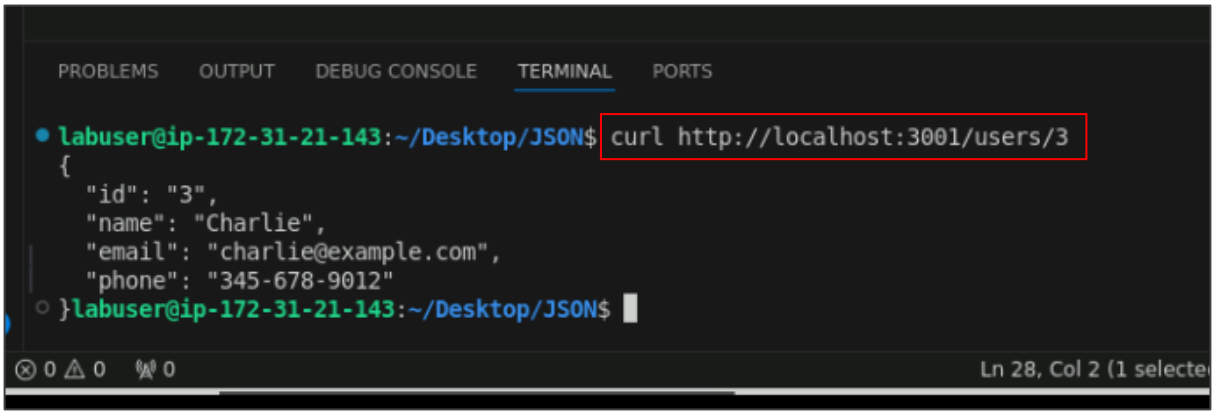
curl -X POST -H "Content-Type: application/json" -d '{"name": "David", "email": "david@example.com", "phone": "456-789-0123"}' http://localhost:3001/users



1.2 Open the **db.json** file in Visual Studio Code to verify that the new user has been successfully added

Step 2: Retrieve data for a specific user (by ID)

2.1 Run the following command to fetch details of a user by their ID:
curl http://localhost:3001/users/3

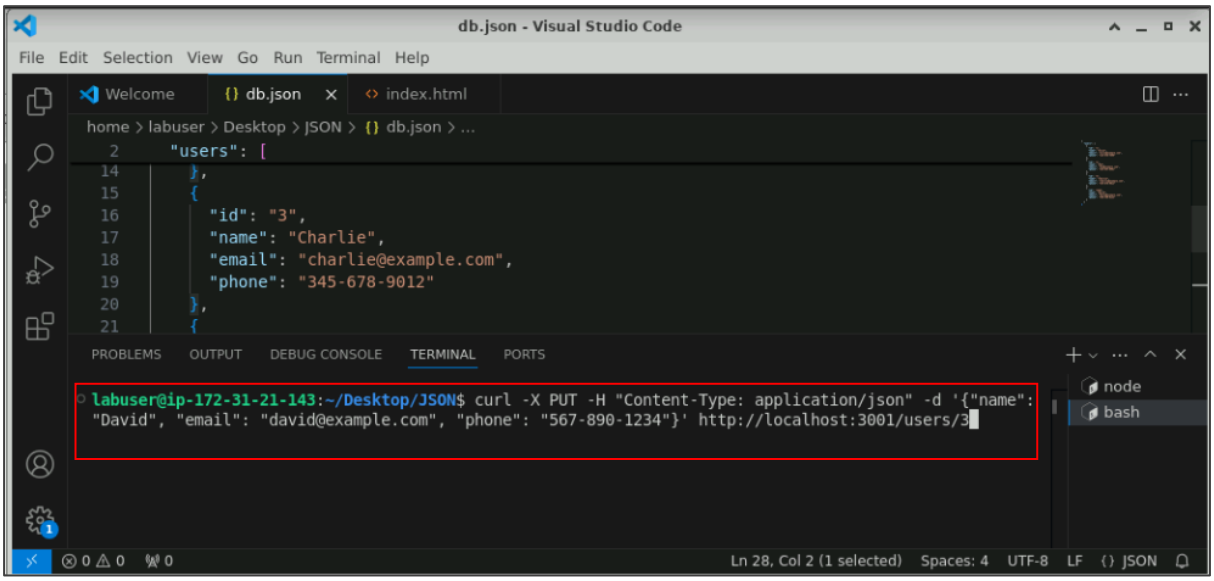


2.2 Compare the output displayed in the terminal with the corresponding user data in the db.json file

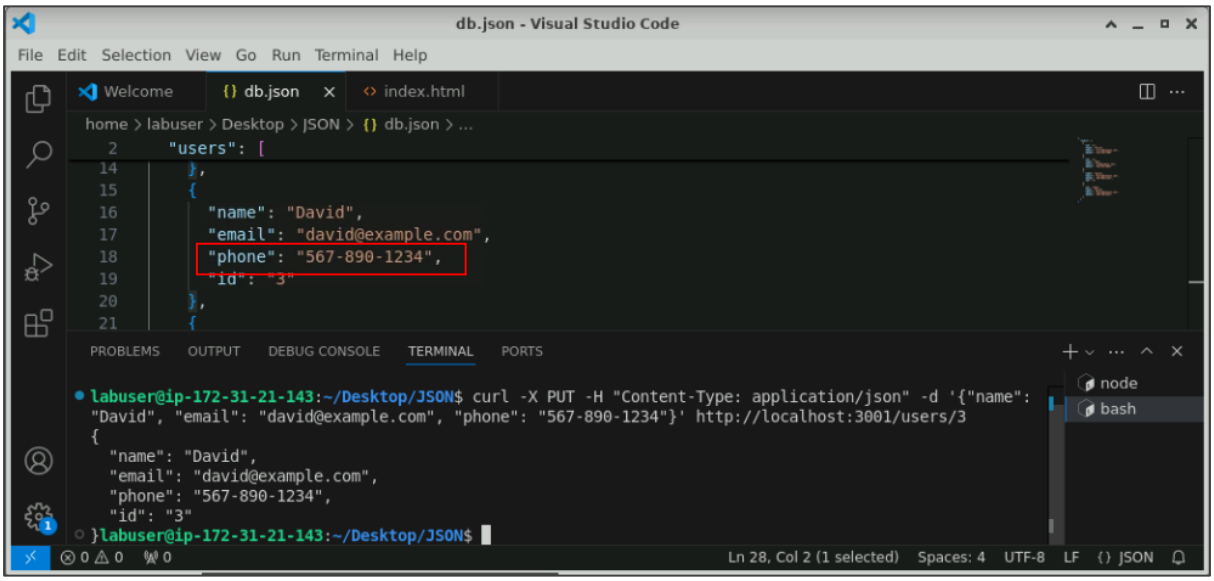
Step 3: Update data for a specific user (by ID)

3.1 Run the following command to update the user's phone number:

curl -X PUT -H "Content-Type: application/json" -d '{"name": "David", "email": "david@example.com", "phone": "567-890-1234"}' http://localhost:3001/users/3

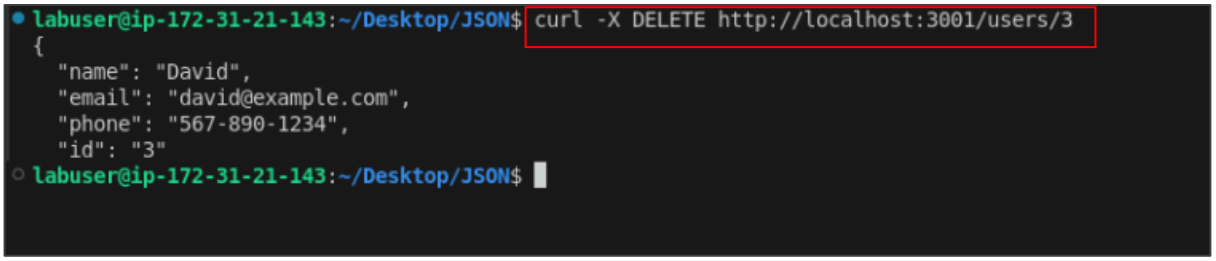


3.2 Open the db.json file and confirm that the user's phone number has been changed



Step 4: Delete a specific user (by ID)

4.1 Run the following command to delete the user with the specified ID:
curl -X DELETE http://localhost:3001/users/3



4.2 Open db.json and verify that the user with ID 3 has been removed

```
home > labuser > Desktop > JSON > {} db.json > [ ] users > {} 2
2     "users": [
3       {
8       },
9       {
10        "id": "2",
11        "name": "Bob",
12        "email": "bob@example.com",
13        "phone": "234-567-8901"
14      },
15      {
16        "id": "7b6d",
17        "name": "David",
18        "email": "david@example.com",
```

By following these steps, you have successfully performed CRUD operations on a JSON database using curl. This process demonstrates how to interact with a REST API efficiently and manipulate data stored in a JSON-based database.